Problem Statement #1: AI-Based Human Behaviour Monitoring Using Pose Estimation and Gesture Analysis

While conducting an examination, continuous manual observation is difficult, leading to situations where suspicious behaviours may go unnoticed or be detected too late. The challenge is to develop an AI-powered video analytics system capable of monitoring seated students in real time using pose estimation and gesture analysis techniques. The system should analyse body posture, head orientation, and hand movements from surveillance camera feeds to identify behaviours that may indicate potential examination misconduct, such as repeated sideward glances, excessive head turning, body rotation toward neighbouring students, unusual hand movements, or attempts to communicate non-verbally. Unlike traditional surveillance systems that rely heavily on facial recognition or manual review of recorded footage, the proposed solution should focus on skeletal keypoints and motion patterns to infer behavioral anomalies while preserving student privacy. The system must operate in real time, track multiple students simultaneously, and generate explainable alerts that highlight specific observed behaviors rather than making direct accusations of cheating.

The system should be robust to variations in seating arrangements, camera angles, lighting conditions, and student demographics while maintaining high accuracy and minimizing false alarms.

Objectives

- Detect and track multiple seated students from live video feeds.
- Extract human pose landmarks and movement patterns.
- Identify suspicious head, body, and hand movements associated with potential examination misconduct.
- Identify objects like Mobile Phone, Paper/Chit or any unwanted objects as may be feasible
- Generate real-time alerts with interpretable explanations.
- Provide behavioral analytics and event logs for invigilator review.
- Ensure privacy-conscious monitoring by avoiding identity-based recognition wherever possible.

Expected Outcome A scalable AI-driven examination monitoring system that assists invigilators in maintaining examination integrity by automatically identifying and reporting unusual behavioral patterns in examination halls through pose estimation and gesture detection.

Risk Factor	Description	Impact
False Positives	Normal student actions (thinking, stretching, looking around) may be incorrectly flagged	Reduced trust in the system and unnecessary invigilator intervention.
False Negatives	Actual suspicious behavior may not be detected.	Potential examination malpractice goes unnoticed.
Occlusion	Students blocking each other from the camera view.	Incomplete pose estimation and tracking errors.
Poor Lighting Conditions	Low light, shadows, or glare affecting video quality.	Reduced detection accuracy.
Camera Blind Spots	Areas not covered by surveillance cameras.	Students in blind spots cannot be monitored effectively.
Head Pose Estimation Errors	Difficulty accurately detecting viewing direction from distant cameras.	Incorrect behaviour classification.
Multi-Person Tracking Failure	Identity switching or loss of student tracking across frames.	Incorrect attribution of suspicious activities.
Network Failure	Loss of connectivity between cameras and processing servers.	Interrupted monitoring and missed events.
Hardware Failure	Camera, GPU, storage, or server malfunction during examinations.	System downtime and loss of monitoring capability.
Privacy Concerns	Student concerns regarding surveillance and data collection.	Regulatory, ethical, and acceptance challenges.